Sreenath Karthick Muthuvel

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SUMMARY

Computer Science undergraduate at IIT Palakkad with research and project experience in machine learning, Explainable AI, LLM systems, MLOps, and software engineering. Experienced in developing deep learning applications, model interpretability tools, agentic LLM workflows, ML pipelines, and software systems. Interested in applied machine learning, trustworthy AI, and ML systems.

EDUCATION

Indian Institute of Technology Palakkad *Aug 2023 – Apr 2027*

B.Tech in Computer Science and Engineering **CGPA: 9.55/10.0**

TOEFL iBT: 110/120 (**CEFR: C1**)
GRE: 317.5/346 (Quantitative Reasoning 165/170)

EXPERIENCE

Research Intern *May 2025 – Jul 2025*

Defence Research and Development Organisation (DRDO: DYSL-CT), Chennai

- Researched Explainable AI techniques including **LIME, SHAP, and Class Activation Maps (CAM)** for interpreting deep learning model predictions.
- Developed **PeliCAM**, a desktop XAI application using **PyTorch** and **PyQt**, supporting CAM-based visualization, LIME explanations, saliency maps, and prediction debugging across **3+ neural network models**.
- Contributed to text classification systems for **English and Indian languages**, gaining experience with practical machine learning workflows.
- GitHub Repository

Software Intern *(Remote)* *Feb 2025 – Dec 2025*

Learning Sciences Research Group, Homi Bhabha Centre for Science Education (HBCSE – TIFR)

- Contributed to an educational cognition research project analyzing the thought processes of **100+ students** learning mathematical concepts.
- Developed interactive web-based tools using JavaScript, HTML, and CSS to support experiments on concepts including proportion, slope, and rate of change.
- Co-authored a peer-reviewed paper published in the *Proceedings of the 33rd International Conference on Computers in Education orgainesed by IIT Madras (ICCE 2025).*
- GitHub Repository
- Link to Conference Paper

Custom Software Engineering Associate *May 2026 – Jul 2026*

Accenture – Coimbatore

- Gained hands-on experience with **Microsoft Azure**, including Virtual Machines, NSGs, RBAC, Azure Policies, Log Analytics, and cloud monitoring.
- Worked with Azure Monitor, Microsoft Defender for Cloud, Microsoft Entra ID, Defender XDR, and Kusto Query Language (KQL).

PROJECTS

SageServe - MLOps for LLM Serving Optimization *GitHub*

- Developed an **MLOps pipeline** around SageServe for workload forecasting and simulation-based optimization of LLM serving infrastructure.
- Implemented data processing, validation, forecasting, model evaluation, and simulation workflows following reproducible **MLOps principles**.
- Integrated **Docker** for reproducible environments and **GitHub Actions** for automated CI workflows.
- *Tools:* Python, MLOps, Docker, GitHub Actions, Pandas, NumPy, Scikit-Learn, Git

Local LLM Debate System for Hallucination Reduction *GitHub*

- Designed a fully local **multi-agent LLM** framework using LangGraph in which agents independently answer, critique, revise, and evaluate responses.
- Built resource routing between **Web Search, RAG, local document retrieval, and Python tools** to support evidence-grounded reasoning.
- Implemented an event-driven monitoring interface using Rich and a thread-safe event bus to visualize routing decisions, evidence, debate rounds, critiques, and final verdicts.
- *Tools:* Python, LangGraph, Ollama, Qwen, LangChain, ChromaDB, HuggingFace Embeddings, Rich, Pandas

Gaur Tool Chain *GitLab*

- Developing a software toolkit for the Foundations of Computing Systems course, involving a custom RISC-V-inspired ISA, assembler, disassembler, CPU simulator and VM Emulator.
- Using **Rocq** to formally reason about the correctness of instruction representations, parsing, assembly, and simulation components.
- *Tools:* OCaml, Rocq, JsofOcaml

Sheet2Slide - Agentic AI Report Generation Framework *GitHub*

- Built a multi-stage **agentic AI pipeline** using LangGraph that transforms structured spreadsheet data into presentation-ready reports.
- Combined structured-output LLM agents with deterministic statistical and visualization tools for data analysis, chart generation, and slide creation.
- Implemented validation and fallback mechanisms to reduce unreliable LLM-generated analysis and keep numerical computation deterministic.
- Automated data profiling, trend detection, correlation analysis, visualization planning, and PowerPoint generation using a local Ollama-hosted LLM workflow.
- *Tools:* Python, LangGraph, Ollama, LangChain, Pandas, Matplotlib, Python-PPTX, Pydantic

COP - Police Workforce Incident Management System *GitHub*

- Developed an incident management system for police workforce operations as part of a team-based software engineering project.
- Implemented application functionality using **C#** and **WinUI**, following software engineering practices for collaborative development.
- Used **Git and GitHub** for version control, collaboration, and project management.
- *Tools:* C#, WinUI, Git, GitHub

Moon Mapping - ISRO Data Analysis *GitHub*

- Analyzed X-ray fluorescence spectra for an ISRO problem statement, implementing data filtering, preprocessing, line-ratio calculations, and visualization on an interactive lunar albedo map.
- Secured **8th position nationally** at Inter-IIT Tech Meet 13.0.
- *Tools:* Python, Pandas, NumPy, SciPy

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, SHAP, LIME, CAM, Pandas, NumPy, SciPy, Matplotlib
- **LLM & AI Systems:** LangGraph, LangChain, Ollama, Qwen, ChromaDB, HuggingFace Embeddings, RAG, Agentic AI
- **MLOps:** MLOps Principles, Docker, GitHub Actions, ML Pipelines, Model Evaluation, Data Validation, Experiment Tracking
- **Programming:** Python, C, C++, C#, OCaml, JavaScript, SQL, Rocq
- **Software & Systems:** Git, GitHub, Linux, REST APIs, Redis, Lua, WinUI
- **Web & Data:** FastAPI, Flask, React, PostgreSQL, HTML, CSS
- **Cloud:** Microsoft Azure, Azure Monitor, Microsoft Defender for Cloud, Microsoft Entra ID, Log Analytics, KQL

ACHIEVEMENTS

- **Inter-IIT Tech Meet 13.0:** Secured **8th position nationally** in the ISRO Problem Statement track.
- **INSPACE Model Rocketry Competition:** Subsystem Lead – Avionics; team secured a **top-15 position nationally**.
- **Core Team Member, NSS IIT Palakkad:** Coordinated and executed social service events on campus.